Supporting Information
When Does Posterior Integration Change Option Values? Evidence from WTI Average Price Options

Anonymous

Scope. This is a separate, anonymized supplement to the Journal of Futures Markets reviewer manuscript. All market price measures are USD per barrel, and Barchart end-of-day `Latest` values are settlement-price proxies rather than independently verified trades or final CME settlement marks. The correlation comparison uses simulated model prices, not historical WTI prices, and does not test calibration to market data.

S1 Full historical-volatility comparison

Table S1 includes the additional earlier-date input benchmarks omitted from the compact main paper. All eight specifications use the same 2,381 contract-date rows over 15 valuation dates. The GARCH parameters are estimated from earlier returns; the valuation-date return updates the conditional variance state. Price benchmarks use the valuation-date CL curve and realized fixings. This is therefore a date-ordered *conditional pricing comparison*, not a fully prospective price forecast.

Table S1: Historical-volatility and earlier option-implied volatility benchmarks; MAE and RMSE in USD/bbl.

Volatility input	Rows	Dates	MAE	RMSE
Long-window historical PI	2,381	15	2.6187	3.4090
Rolling 252 returns	2,381	15	5.2183	6.3933
Rolling 126 returns	2,381	15	8.3419	9.7846
Rolling 63 returns	2,381	15	4.5201	5.6465
EWMA, half-life 63 business days	2,381	15	5.6616	6.8824
Horizon-matched GARCH(1,1)	2,381	15	2.6730	3.4441
Earlier-date expanding APO smile	2,381	15	0.1088	0.1725
Previous-day APO smile	2,381	15	0.1075	0.1594

S2 Numerical agreement and simulation-based calibration

Table S2 compares PI–PM differences for selected empirical conditions across alternative numerical pricing methods. Agreement on the *difference* does not imply equality of absolute Curran and exact arithmetic-average payoff prices. The Gaussian posterior simulation-based calibration uses 200,000 prior-predictive datasets in total, with 50,000 at each historical length. These checks assess the statistical inference code, not the accuracy of the exchange-price comparisons.

A separate posterior-RMS plug-in sensitivity replaces mean volatility by $\sqrt{\mathrm{E}[\sigma^2 \mid \mathcal{D}]}$ under the same Curran approximation. This is a distinct point-estimator benchmark, not posterior price integration.

Table S2: Representative numerical PI–PM differences (USD/bbl).

October call, strike 98	Pseudo-MC	Curran	Sobol
2026-08-27	0.0013683	0.0013690	0.0013717
2026-08-28	0.0013394	0.0013402	0.0013393

Table S2b: Gaussian volatility posterior simulation-based calibration.

Historical n	Mean rank	KS p	Empirical 95% coverage
21	0.5000	0.861	0.9500
63	0.4993	0.348	0.9505
252	0.4978	0.210	0.9483
1,260	0.5027	0.106	0.9505

S3 Liquidity, timing and data provenance

The 2,381-row pricing panel includes 2,319 observations with zero recorded trading volume. Only 62 observations on 13 valuation dates have positive reported volume. Implied volatility still gives lower pricing errors in this restricted sample, but there are too few trades to draw general conclusions about prices available for execution. Valuation-date cluster resampling accounts for cross-contract dependence within a date but not all serial dependence across nearby dates.

Table S3: Positive-volume conditional repricing comparison.

Volatility input	Rows	Dates	MAE	RMSE
Historical PI	62	13	1.6340	2.3391
Earlier-date expanding APO smile	62	13	0.1003	0.1358
Previous-day APO smile	62	13	0.1084	0.1333

Vendor first-publication timestamps and record revisions have not been established. CME Clearing Advisory 20-198, dated 15 May 2020, specifies Curran for WTI Average Price Option end-of-day valuation effective 18 May 2020. The extent to which the vendor `Latest` marks reproduce the exchange's model-generated settlements remains unverified. See `https://www.cmegroup.com/notices/clearing/2020/05/Chadv20-198.pdf`. If the price proxy and our model share a valuation method, their agreement may partly reflect that method rather than an independent market test.

S4 External vanilla-WTI implied volatility

The cross-instrument comparison estimates volatility from earlier LO futures-option prices and applies those estimates to APO fixing dates. We first combine the LO volatilities using a root-mean-square average weighted by the number of remaining fixings. As an alternative, we choose a constant volatility that matches the variance of the remaining average under the one-factor model. Results in Table S4 apply to 253 contract-date APO observations over 10 valuation dates.

S5 Numerical reference and volatility-grid sensitivity

The synthetic comparison uses 160 independent return histories and 480 option observations at each assumed futures correlation. Each return history is used for three strikes. The volatility posterior and PI/PM prices assume $\rho = 1$ throughout. For each assumed correlation $\rho \in \{1, 0.75, 0.5\}$, we independently estimate the call

Table S4: Combining LO implied volatilities for APO pricing (USD/bbl).

Specification	MAE	RMSE
Previous-day LO, weighted RMS	0.1559	0.2599
Previous-day LO, variance-matched	0.1562	0.2612
Expanding LO, weighted RMS	0.1604	0.2858
Expanding LO, variance-matched	0.1611	0.2869

Table S4b: Absolute changes in LO implied volatility estimated with successively finer Cox–Ross–Rubinstein trees (156 contract-date observations).

Steps	Mean	Median	95th pct.	Maximum
80 to 160	0.000820	–	–	–
160 to 320	0.000402	0.000284	0.001089	0.001810

price at the true volatility using scrambled Sobol simulation and an analytically priced geometric-average control variate. These are hypothetical correlations, not empirical estimates for the WTI futures curve.

We repeated the posterior calculation with 1,025, 2,049, and 4,097 volatility-grid points, using the same historical summaries and QMC reference prices as in the original 513-point calculation. The number of reference prices inside the posterior intervals is unchanged across all four grids. For one case with $\rho = 0.75$, the QMC uncertainty interval overlaps a posterior interval boundary on the finer grids. The absolute posterior quantile endpoints nevertheless change by up to 0.00523 USD/bbl between 513 and 4,097 points. Thus, unchanged coverage counts do not establish accuracy better than 0.001 USD/bbl for the interval endpoints.

Table S5: Nominal 95% posterior model-price interval coverage, number covered among 480 option-state rows at each assumed reference correlation. The same historical returns are reused.

Volatility nodes	$\rho = 1.00$	$\rho = 0.75$	$\rho = 0.50$
513	456 (95.0%)	313 (65.2%)	153 (31.9%)
1,025	456 (95.0%)	313 (65.2%)	153 (31.9%)
2,049	456 (95.0%)	313 (65.2%)	153 (31.9%)
4,097	456 (95.0%)	313 (65.2%)	153 (31.9%)

Table S5b: Maximum absolute posterior grid differences against the 4,097-node benchmark (USD/bbl); this is not a proof of the exact continuous-grid limit.

Nodes	PI	PM	2.5% endpoint	97.5% endpoint
513	4.50×10^{-9}	0.0000556	0.005227	0.004029
1,025	1.08×10^{-9}	0.0000152	0.001215	0.001028
2,049	2.15×10^{-10}	0.00000496	0.000275	0.000244

The observations are not independent across strikes, and the study was not preregistered. These are frequentist repeated-sampling checks of posterior *model-price* intervals against known parameters, not observed price intervals, executable performance, or evidence of real-market correlation misspecification. The study can be reproduced using `experiments/jfm_interval_grid_convergence.py`, with the stored results and QMC reference inputs.